

- 1 In the roots of plants, specialised tissues are formed from a region in the root tip known as the root apical meristem.

Fig. 1.1 shows the different regions of a root tip.

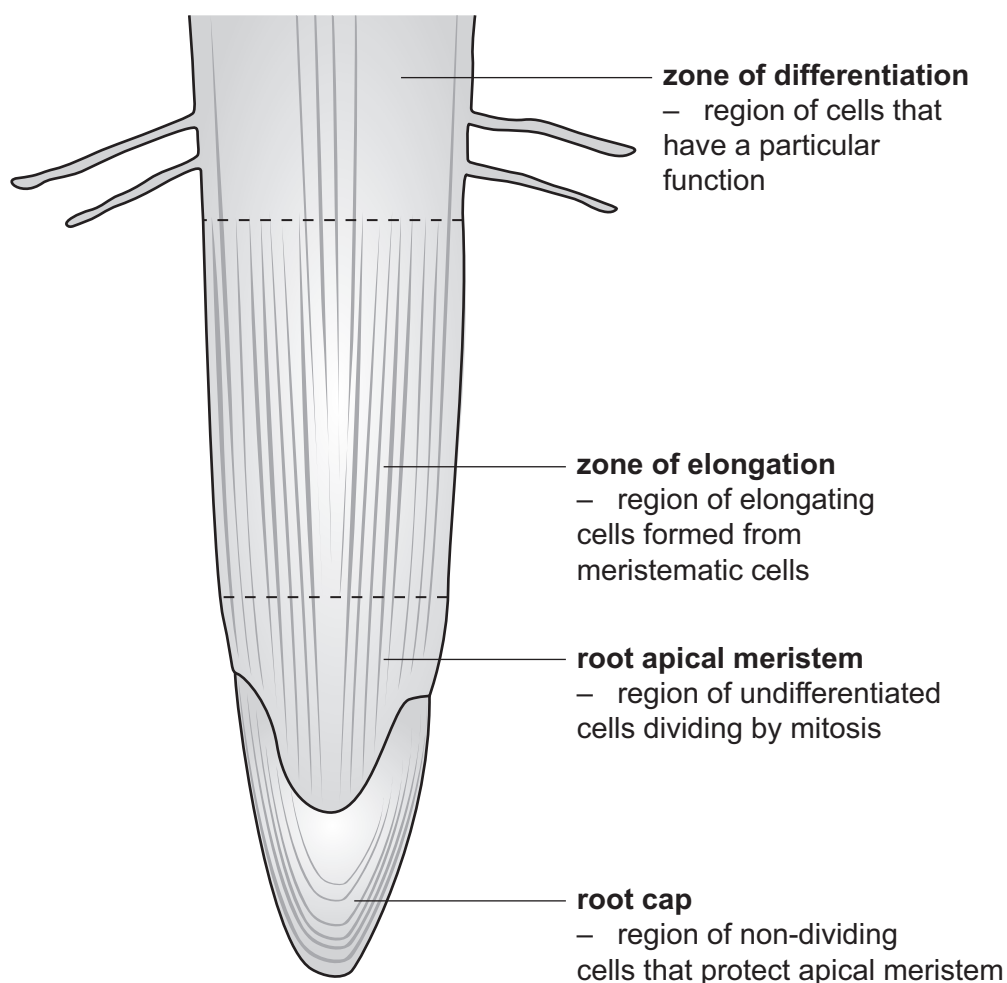


Fig. 1.1

- (a) Cells in all stages of mitosis will be visible in the root apical meristem by using the high power of a light microscope.

- (i) Name the stage of mitosis during which the chromosomes are arranged at the spindle equator of the cell.

..... [1]

- (ii) Early in mitosis, the nuclear envelope breaks up into vesicles and so will not be seen in some cells in the root apical meristem.

Name the stage of mitosis during which the vesicles fuse to form new nuclear envelopes.

..... [1]



(b) In the zone of elongation, shown in Fig. 1.1, the newly formed cells expand and elongate.

Two main events that occur in these cells are:

- the entry of water into the cells
- the formation of the large vacuole.

(i) Outline the process by which water enters the cells.

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(ii) Outline the structure of a fully mature plant vacuole.

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..... [2]

(c) In the zone of differentiation, shown in Fig. 1.1, two types of vascular (transport) tissue form: xylem and phloem.

Draw a diagram showing the distribution of the vascular tissue as seen in a transverse section of a root in the zone of differentiation.



- (d) Starch can be stored in the roots of plants. This can be converted to sucrose, which is loaded into phloem sieve tubes for transport within the phloem sap to growing areas in the stem.

After loading of sucrose into phloem sieve tubes, the transport of phloem sap occurs by mass flow.

Explain how the phloem sap is moved within the sieve tubes by mass flow from the root storage areas to the growing areas of the stem.

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[Total: 13]